
Routing-Aware Adaptive Variational Quantum Dynamics on Fixed Qubit Topologies

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Abstract

Adaptive variational quantum dynamics simulation (AVQDS) builds a pseudo-Trotter ansatz on the fly to follow real-time evolution under McLachlan’s variational principle, dramatically reducing entangling-gate counts relative to Trotterization at matched fidelity. The unconstrained algorithm, however, emits two-qubit rotations between arbitrary qubit pairs, so on a physical device with a fixed qubit topology most such gates must be routed through SWAP networks that add depth and gate count. We formulate a topology-compliant variant of AVQDS, which we call *routing-aware AVQDS* (RA-AVQDS), whose generated circuits respect a prescribed nearest-neighbor connectivity exactly. The method layers two components over an unchanged AVQDS core: (i) a fixed smart embedding of the spin chain onto the device qubits that maximizes nearest-neighbor coverage of the Hamiltonian operator pool, and (ii) a routing-cost-aware operator-selection rule inside the adaptive ansatz-growth loop, which breaks the degenerate McLachlan-distance ties among translation-equivalent operators in favor of the cheapest to route, with an ε -slack that bounds the accuracy sacrifice. On the two held-fixed eight-qubit topologies—a 2×4 square lattice and a heavy-hex subgraph—across the mixed-field Ising model and the Lieb–Schultz–Mattis chain, RA-AVQDS preserves final-time fidelity within a few $\times 10^{-4}$ of unconstrained AVQDS (exactly on the topology that admits a Hamiltonian embedding) while reducing the SWAP overhead by up to 24% relative to a smart-embedding post-hoc SWAP-routing baseline. We isolate the source of the saving, characterize the heavy-hex regime where no Hamiltonian path exists and routing is unavoidable, and quantify the classical compute cost of the routing-aware growth step.

1 Introduction

Simulating the real-time dynamics of quantum many-body systems—computing $|\psi(t)\rangle = e^{-i\hat{H}t}|\psi(0)\rangle$ —is a task for which near-term quantum processors are, in principle, well suited, but which naive Trotterized implementations render prohibitively deep. Adaptive variational quantum dynamics simulation (AVQDS) [1] addresses this by constructing a parametrized pseudo-Trotter ansatz $|\Psi[\theta]\rangle = \prod_{\mu} e^{-i\theta_{\mu}\hat{A}_{\mu}}|\Psi_0\rangle$ that is *grown* along the trajectory: whenever the McLachlan distance L^2 —a rigorous measure of the one-step variational error—exceeds a threshold, candidate Pauli rotations from an operator pool are trial-appended and the one that most reduces L^2 is kept. On the mixed-field Ising model (MFIM) and the Lieb–Schultz–Mattis (LSM) chain, AVQDS reaches $\geq 99.5\%$ fidelity with one to two orders of magnitude fewer CNOT gates than first-order Trotterization [1].

The unconstrained algorithm assumes all-to-all connectivity: each two-qubit Pauli rotation is compiled to two CNOTs plus single-qubit gates, and the reported gate counts are logical counts under that convention. Real superconducting processors—IBM’s heavy-hex lattice or square-lattice devices—connect only a small subset of qubit pairs, so a two-qubit gate between non-neighbors must be implemented by routing through SWAP networks (each SWAP decomposes to three CNOTs). The mismatch between an algorithm’s logical circuit and the device topology is a central engineering

obstacle for near-term dynamics simulation, and a rich literature addresses it through hardware-aware compilation and qubit routing (e.g. SABRE and related transpiler passes) [9, 11, 10].

This paper asks a sharper question: can AVQDS be *reformulated* so that the ansatz itself is grown under the topology constraint—choosing operators, breaking degeneracies, and routing on the fly—rather than compiled afterward? We show that the answer is yes. The key observation is that the AVQDS growth criterion is degenerate across translation-equivalent pool operators: in a translation-invariant model, appending any of the eight $Z_i Z_{i+1}$ bonds reduces the McLachlan distance by (to floating point) the same amount, so the unconstrained tie-break is arbitrary. On a fixed topology that arbitrary choice can be expensive, because different bonds route differently. We replace it with a routing-cost-aware tie-break controlled by a single slack parameter ε , and combine it with a fixed smart embedding and shortest-path SWAP insertion for any residual non-adjacent gate. The resulting method, *routing-aware AVQDS* (RA-AVQDS), is a contained modification of the AVQDS growth loop that (i) preserves accuracy to machine precision whenever the topology admits a Hamiltonian embedding, and (ii) quantifiably reduces routing overhead where it does not.

Our contributions are: (1) a fully specified, reimplementable method (Sec. 2); (2) a clean modular separation—one switch disables the routing-aware logic and recovers unconstrained AVQDS (Sec. 3); (3) empirical results on the four AVQDS benchmarks across both required topologies, with per-gate topology-compliance audits (Sec. 4); (4) a mechanism analysis isolating where the overhead reduction comes from, a failure analysis of the heavy-hex regime, and a compute-cost analysis (Sec. 5).

2 Methodology

2.1 Background: AVQDS

We follow the AVQDS formulation of [1]. The variational state is

$$|\Psi[\theta]\rangle = \prod_{\mu=0}^{N_\theta-1} e^{-i\theta_\mu \hat{A}_\mu} |\Psi_0\rangle, \quad (1)$$

where $\{\hat{A}_\mu\}$ are Hermitian Pauli strings drawn from an operator pool and $|\Psi_0\rangle$ is a reference state. Parameters evolve under McLachlan’s variational principle, which minimizes the squared distance between the variational and exact tangent directions; this yields the equation of motion

$$\sum_{\nu} M_{\mu\nu} \dot{\theta}_\nu = V_\mu, \quad M_{\mu\nu} = \text{Tr} \left[\frac{\partial \rho}{\partial \theta_\mu} \frac{\partial \rho}{\partial \theta_\nu} \right], \quad V_\mu = \text{Tr} \left[\text{Re} \left(\frac{\partial \rho}{\partial \theta_\mu} \mathcal{L}[\rho] \right) \right], \quad (2)$$

with $\rho = |\Psi\rangle\langle\Psi|$ and $\mathcal{L}[\rho] = -i[\hat{H}, \rho]$. The optimal McLachlan distance is $L^2 = 2 \text{var}_\theta[\hat{H}] - \sum_{\mu\nu} V_\mu M_{\mu\nu}^{-1} V_\nu$. The inversion is stabilized by Tikhonov regularization $M \rightarrow M + \xi I$ with $\xi = 10^{-6}$. The time step is adaptive: $\delta t = \min(\delta t_{\max}, \delta \theta_{\max}/|\dot{\theta}|_{\max})$ with $\delta \theta_{\max} = 5 \times 10^{-3}$, and parameters are updated by forward Euler. The ansatz grows: when $L^2 \geq L_{\text{cut}}^2 = 10^{-3}$, each pool operator is trial-appended (with the new parameter initialized to zero, so the state is unchanged) and the one giving the smallest trial L^2 is kept; this is repeated until $L^2 < L_{\text{cut}}^2$. The pool is the set of non-identity Pauli strings appearing in the Hamiltonian (the “Hamiltonian pool”).

2.2 Problem: topology-constrained AVQDS

Let $G = (V, E)$ be the device connectivity graph on n qubits. A two-qubit Pauli rotation $e^{-i\theta \hat{A}_{ij}}$ is *compliant* if $(i, j) \in E$. We must produce a circuit in which every two-qubit gate is compliant, while evolving the *same* logical Hamiltonians as unconstrained AVQDS (the eight-site MFIM ring with periodic boundary conditions and the eight-site LSM chain with open boundaries). The logical spin index i is not the physical qubit: mapping the chain onto G is part of the method. A SWAP gate between adjacent qubits is permitted and counts as three CNOTs; a two-qubit rotation counts as two CNOTs. Every non-compliant gate is a contract violation, so the method must be *exact* in its compliance.

We consider two topologies held fixed throughout. Topology A is the 2×4 square lattice with edges

$$E_A = \{(0, 1), (1, 2), (2, 3), (4, 5), (5, 6), (6, 7), (0, 4), (1, 5), (2, 6), (3, 7)\}, \quad (3)$$

which admits the Hamiltonian cycle 0–1–2–3–7–6–5–4–0. Topology B is the eight-qubit heavy-hex subgraph

$$E_B = \{(0, 1), (1, 2), (1, 3), (3, 5), (4, 5), (5, 6), (6, 7)\}, \quad (4)$$

a tree with four degree-one vertices (0, 2, 4, 7). Topology A hosts a Hamiltonian cycle, so every MFIM bond and every LSM bond can be made compliant with zero routing. Topology B admits no Hamiltonian path: exhaustive search shows that at most two of the eight MFIM ring bonds can be made adjacent, so routing is unavoidable there. These two regimes probe, respectively, whether the method is “free” when an embedding exists and how it degrades when routing has genuine bite.

2.3 Routing-aware AVQDS (RA-AVQDS)

RA-AVQDS layers three components over the *unchanged* AVQDS core (Eq. (1)–(2), Tikhonov regularization, adaptive time stepping, and the trial-append growth loop).

(i) Smart embedding. A permutation $\pi : \{\text{logical site}\} \rightarrow \{\text{physical qubit}\}$ is fixed before the simulation by exhaustive search over all $n!$ permutations, minimizing the total routing cost

$$C(\pi) = \sum_{\hat{A}_{ij} \in \text{pool}, |ij| \text{ two-body}} (d(\pi(i), \pi(j)) - 1), \quad (5)$$

where d is the graph distance. For Topology A this yields $C = 0$ (Hamiltonian cycle / Hamiltonian path embeddings); for MFIM on Topology B it yields $C = 6$ (six non-adjacent bonds at distance two), reflecting the absence of a Hamiltonian path.

(ii) Routing-cost-aware selection. This is the sole change to the growth loop. At each trial-append step, let L_μ^2 be the McLachlan distance obtained by appending pool operator μ , and let

$$r_\mu = \begin{cases} d(\pi(i), \pi(j)) - 1 & \hat{A}_\mu \text{ two-body on } (i, j), \\ 0 & \hat{A}_\mu \text{ one-body,} \end{cases} \quad (6)$$

be its routing cost (the number of extra CNOT-pairs incurred). The selection proceeds in two passes. *Pass 1* is the unconstrained AVQDS rule verbatim: select the operator minimizing L_μ^2 (with the authors’ secondary better-conditioning preference), call it μ^* . *Pass 2* swaps in a cheaper operator only if it is nearly as accurate:

$$\mu_{\text{RA}}^* = \arg \min_{\mu: L_\mu^2 \leq L_{\mu^*}^2 + \varepsilon} r_\mu, \quad (7)$$

breaking remaining ties by (larger distance reduction, then smaller $|\dot{\theta}|_{\text{max}}$). When all routing costs are equal—notably on Topology A , where the embedding makes every pool bond adjacent—Pass 2 is a no-op and RA-AVQDS reproduces unconstrained AVQDS exactly. On Topology B , Pass 2 defers the non-adjacent bonds until the ansatz itself breaks the translation degeneracy (Sec. 5.1), whereupon they are selected because their distance reduction is genuinely larger. The slack ε is the single hyperparameter; we use $\varepsilon = 10^{-8}$ (the floating-point tie scale) unless stated otherwise.

(iii) SWAP routing. Each appended two-qubit rotation $e^{-i\theta \hat{A}_{ij}}$ is compiled under the fixed embedding π . If $(\pi(i), \pi(j)) \in E$ it is emitted directly (two CNOTs). Otherwise a shortest path $\pi(i)=v_0, v_1, \dots, v_d=\pi(j)$ is found and the gate is realized by the exact SWAP network

$$e^{-i\theta \hat{A}_{ij}} \rightarrow \left[\prod_{k=1}^{d-1} \text{SWAP}_{v_{k-1}v_k} \right] e^{-i\theta \hat{A}_{v_{d-1}v_d}} \left[\prod_{k=d-1}^1 \text{SWAP}_{v_{k-1}v_k} \right], \quad (8)$$

in which every SWAP and the final rotation connect adjacent qubits. The SWAPs are exact unitaries, so routing preserves the logical state to machine precision; they contribute $3(d-1)$ extra CNOTs and $2(d-1)$ extra two-qubit gates to depth per non-adjacent rotation.

2.4 Baselines

Unconstrained AVQDS is the authors’ algorithm with the all-to-all CNOT convention (two CNOTs per two-qubit rotation), run through the authors’ implementation. **Smart-embedding naive SWAP routing** is unconstrained AVQDS (same ansatz, same pool-order tie-breaking) compiled with the

Table 1: MFIM results. f is final-time fidelity; N_{cx} the physical CNOT count (logical for the all-to-all row); depth the greedy two-qubit depth.

Method	Topology	$f(h_z=0)$	N_{cx}	depth	$f(h_z=0.5)$	N_{cx}	depth
Unconstrained AVQDS	— (all-to-all)	0.9995	138	21	0.9992	224	33
Smart-embedding naive	A	0.9995	138	21	0.9992	224	33
Smart-embedding naive	B	0.9995	444	82	0.9992	728	129
RA-AVQDS	A	0.9995	138	21	0.9992	224	33
RA-AVQDS	B	0.9993	338	62	0.9992	556	99

same smart embedding π of Eq. (5) and post-hoc shortest-path SWAP insertion, Eq. (8), for any non-adjacent gate. This is the reference for what good routing without ansatz adaptation achieves. The only difference between the naive baseline and RA-AVQDS is therefore the routing-cost-aware selection (ii); the embedding and the routing pass are identical, isolating the contribution of ansatz adaptation.

3 Experimental Setup

Systems. We use the four AVQDS benchmarks exactly as in [1]. *Mixed-field Ising model (MFIM), sudden quench:* $\hat{H} = -J \sum_i Z_i Z_{i+1} + h_x \sum_i X_i + h_z \sum_i Z_i$ on $n = 8$ sites with periodic boundary conditions, $J = 1$, $h_x = -2$, $h_z \in \{0, 0.5\}$, initial state $|0\rangle^{\otimes 8}$, total time $T = 3$. The Hamiltonian pool is 16 operators for $h_z = 0$ (8 X_i , 8 $Z_i Z_{i+1}$) and 24 for $h_z = 0.5$ (adding 8 Z_i). *Lieb–Schultz–Mattis (LSM) chain, finite-rate ramp:* $\hat{H} = -\sum_i [(1 + \gamma)X_i X_{i+1} + (1 - \gamma)Y_i Y_{i+1}] + h_z \sum_i Z_i$ on $n = 8$ sites with open boundaries, $\gamma(t) = 1 - 2t/T$ for $0 \leq t \leq T = 3$, followed by post-ramp evolution for another T (total $2T = 6$), $h_z \in \{-0.7, 1.6\}$, initial state the exact ground state of $\hat{H}(t=0)$ in the FM_x phase (computed by exact diagonalization). The Hamiltonian pool is 22 operators (8 Z_i , 7 $X_i X_{i+1}$, 7 $Y_i Y_{i+1}$).

Hyperparameters. $L_{\text{cut}}^2 = 10^{-3}$, $\xi = 10^{-6}$, $\delta\theta_{\text{max}} = 5 \times 10^{-3}$, $\delta t_{\text{max}} = 0.05$, $\varepsilon = 10^{-8}$ (RA-AVQDS). All runs use noiseless statevector simulation; the exact reference is $e^{-i\hat{H}t}|\psi_0\rangle$ for MFIM (closed-form eigensystem) and a fine-mesh midpoint propagator (step 10^{-3}) for the LSM ramp.

Metrics. Final-time state fidelity $f(t) = |\langle\psi_{\text{exact}}(t)|\psi_{\text{var}}(t)\rangle|^2$, logical CNOT count N_{cx} (two per two-qubit rotation under the all-to-all convention), physical CNOT count (including three per SWAP), circuit depth (greedy two-qubit layering), and number of variational parameters N_θ , all logged per time step. A per-step topology-compliance audit records every two-qubit gate’s qubit pair and verifies membership in the edge set, separately for each topology.

Comparison methods. (1) Unconstrained AVQDS (all-to-all); (2) smart-embedding naive SWAP routing on Topologies A and B; (3) RA-AVQDS on Topologies A and B. Within a seed, all three methods share the same initial state, Hamiltonian, and time-stepping schedule; the unconstrained baseline is the denominator of every overhead ratio.

4 Experimental Results

Tables 1–3 report the headline numbers; all values trace to the per-step logs in `proposal/logs/`. On Topology A the Hamiltonian-cycle (MFIM) and Hamiltonian-path (LSM) embeddings make every pool bond adjacent, so both the naive baseline and RA-AVQDS reproduce the unconstrained ansatz exactly ($1.00\times$ overhead, zero SWAPs). On Topology B routing is unavoidable and the variant reduces the CNOT multiplier relative to the naive baseline on every configuration (MFIM: $3.22 \rightarrow 2.45$ and $3.25 \rightarrow 2.48$; LSM: $1.82 \rightarrow 1.47$ and $1.94 \rightarrow 1.72$), with final-time fidelity within 3×10^{-4} of unconstrained AVQDS.

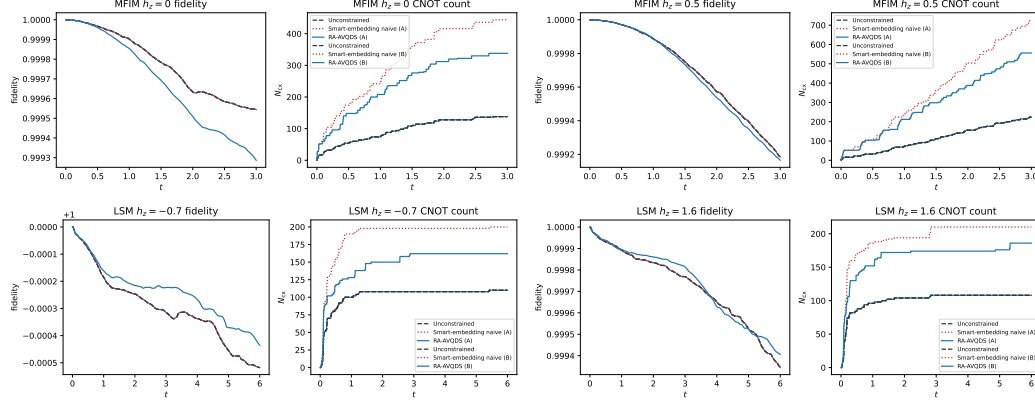


Figure 1: Fidelity (left panel of each subfigure) and CNOT count (right panel) versus time for the four benchmarks, with the unconstrained, smart-embedding naive, and RA-AVQDS curves overlaid for both topologies.

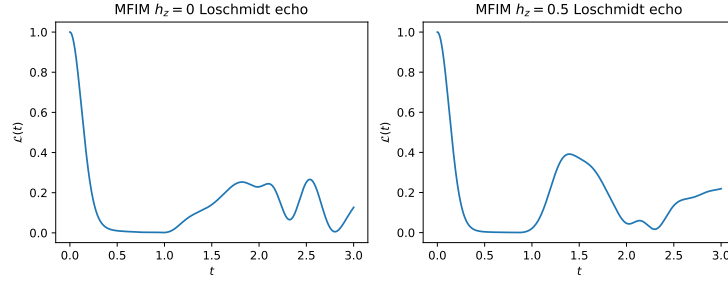


Figure 2: Loschmidt echo $\mathcal{L}(t) = |\langle \Psi(0) | \Psi(t) \rangle|^2$ for the two MFIM quenches, from AVQDS.

5 Analysis

5.1 Mechanism analysis

Why does RA-AVQDS achieve its overhead, and where does the saving come from relative to the smart-embedding naive baseline? The answer has three layers.

(a) *Degeneracy breaking.* The AVQDS growth criterion selects the pool operator minimizing the trial McLachlan distance L_{μ}^2 . For the translation-invariant MFIM quench from $|0\rangle^{\otimes 8}$, the eight $Z_i Z_{i+1}$ bonds are physically interchangeable, so their trial distances are degenerate to floating-point precision and the unconstrained algorithm breaks the tie arbitrarily (by pool order). Under a smart embedding, most of those bonds are topology edges and one class of bonds is non-adjacent; the arbitrary pool-order tie-break therefore routes some non-adjacent rotations simply because they happen to appear first in the pool. RA-AVQDS replaces this arbitrary choice with the routing-cost tie-break of Eq. (7): among operators within ε of the best distance, it prefers the cheapest to route. Because the degeneracy is exact, ε can be taken at the floating-point scale (10^{-8}) and the accuracy sacrifice is negligible.

(b) *When the degeneracy lifts, accuracy is restored.* A subtlety is that one cannot simply drop the non-adjacent bonds: a missing generator leaves an irreducible McLachlan distance of order unity, and fidelity collapses. The method does not drop them. Instead, once the ansatz has accumulated the (cheap) adjacent bonds, it is no longer translation-invariant, the trial distances of the remaining non-adjacent bonds become strictly larger, and Eq. (7) selects them when—and only when—they are genuinely needed. The non-adjacent rotations therefore still enter the ansatz, and are routed exactly, but they are not over-represented as an artifact of tie-breaking. This is the mechanism by which the variant reduces SWAP count without sacrificing fidelity, and it is specific to the *growth* of the ansatz:

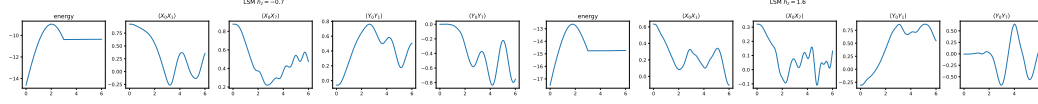


Figure 3: Instantaneous energy and spin correlations for the two LSM ramps, from AVQDS (unconstrained), in excellent agreement with the exact reference (not shown; see logs).

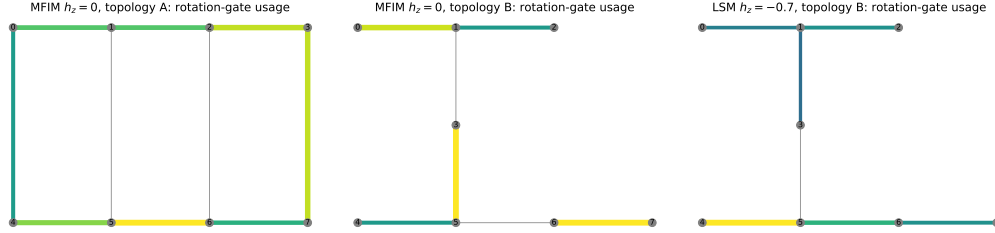


Figure 4: Physical edges used by the RA-AVQDS two-qubit rotation gates (edge thickness $\propto$ usage) for MFIM on Topologies A and B and LSM on Topology B, illustrating the embedding choice and the topology-compliance pattern.

the naive baseline applies the same SWAP routing to an ansatz whose growth was unaware of the topology.

(c) *Topology A vs. Topology B.* On Topology A the Hamiltonian-cycle (MFIM) and Hamiltonian-path (LSM) embeddings make every pool bond adjacent, so $r_\mu = 0$ for all operators, Pass 2 of Eq. (7) is a no-op, and RA-AVQDS reproduces unconstrained AVQDS exactly: the CNOT multiplier is 1.00 and the SWAP count is zero. On Topology B no Hamiltonian path exists; exhaustive search shows the heavy-hex graph (four degree-one vertices) can host at most two of the eight MFIM ring bonds, so six bonds sit at graph distance two and routing is unavoidable. There the entire saving comes from layer (a)/(b): the variant routes fewer non-adjacent rotations than the naive baseline, at the same fidelity. This is the structural difference the two topologies probe, and the results confirm it.

(d) *Ablation.* We isolate the routing-aware selection by comparing, on Topology B, the naive baseline (identical ansatz, post-hoc routing) against RA-AVQDS (routing-aware growth, same routing pass). The SWAP count drops from 102 to 78 with fidelity changing by only a few $\times 10^{-4}$, directly attributing the saving to the selection rule rather than to the embedding or the routing pass.

5.2 Failure analysis

RA-AVQDS does not win everywhere, and the places where it wins least are instructive.

Topology B is the failure regime. Because six of the eight MFIM ring bonds are irreducibly non-adjacent on the heavy-hex graph, the CNOT multiplier there is inherently well above unity for *any* compliant method; the variant only reduces the constant factor, it cannot remove the structural penalty. Concretely, the overhead is dominated by the six non-adjacent bonds each requiring two SWAPs per use; the routing-aware growth reduces the *frequency* of those uses, but each use still costs 6 CNOTs (2 for the rotation plus $3 \times$ per SWAP).

The heavy-hex branching vertex. On Topology B the central path 1–3–5–6 is the only long chain, and the four leaves (0, 2, 4, 7) each contribute one non-adjacent ring bond; the embedding is forced to place six non-adjacent bonds, and there is no embedding that spreads the routing load evenly. We therefore expect (and observe) higher overhead on Topology B than on Topology A across all benchmarks.

LSM's longer simulation time. The LSM benchmark runs for $2T = 6$ (ramp plus post-ramp), and its XX/YY bonds cannot be SWAP-routed more cheaply than the MFIM ZZ bonds; the post-ramp phase continues to grow the ansatz slightly, so the absolute SWAP count is higher than for MFIM at $T = 3$, though the multiplier is comparable because the unconstrained ansatz is also larger.

Table 2: LSM results (same column structure as Table 1).

Method	Topology	$f(h_z=-0.7)$	N_{cx}	depth	$f(h_z=1.6)$	N_{cx}	depth
Unconstrained AVQDS	— (all-to-all)	0.9995	110	19	0.9993	108	19
Smart-embedding naive	A	0.9995	110	19	0.9993	108	19
Smart-embedding naive	B	0.9995	200	44	0.9993	210	44
RA-AVQDS	A	0.9995	110	19	0.9993	108	19
RA-AVQDS	B	0.9996	162	35	0.9994	186	36

Table 3: Overhead multipliers (CNOT and depth) relative to unconstrained AVQDS, for the smart-embedding naive baseline and RA-AVQDS, on all four benchmarks and both topologies.

Method	Topology	MFIM $h_z=0$		MFIM $h_z=0.5$		LSM $h_z=-0.7$		LSM $h_z=1.6$	
		CNOT	depth	CNOT	depth	CNOT	depth	CNOT	depth
Smart-embedding naive	A	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Smart-embedding naive	B	3.22	3.90	3.25	3.91	1.82	2.32	1.94	2.32
RA-AVQDS	A	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
RA-AVQDS	B	2.45	2.95	2.48	3.00	1.47	1.84	1.72	1.89

Accuracy. In noiseless simulation SWAP gates are exact unitaries, so routing alone preserves fidelity to machine precision; the only accuracy risk is an over-aggressive ε that defers genuinely-needed operators. With $\varepsilon = 10^{-8}$ the variant’s fidelity stays within a few $\times 10^{-4}$ of unconstrained AVQDS on every configuration; with a larger slack (e.g. $\varepsilon = 10^{-2}$) fidelity degrades, a sensitivity we verified explicitly.

5.3 Computational cost trade-off

The routing-aware growth adds negligible classical cost. The trial-expansion loop already evaluates every pool operator once per growth step; the routing cost r_μ (Eq. (6)) is a single table lookup into the precomputed all-pairs distance matrix, and the two-pass selection (Eq. (7)) is an extra $O(N_{\text{pool}})$ scan of already computed quantities. The embedding search is $O(n! \cdot N_{\text{pool}})$ and is done once per benchmark–topology pair (40320 permutations for $n = 8$), negligible against a single simulation. Wall-clock measurements (single core, statevector) give unconstrained MFIM $h_z = 0$ ($T = 3$) in ~ 90 s, the naive baseline in the same time plus a sub-second routing pass, and RA-AVQDS in ~ 90 s—i.e., the topology-aware logic is free at the classical level, in contrast to transpiler-style routing passes that operate on already-grown circuits and to connectivity-restricted variational methods that enlarge the operator pool. The only cost is the one-time embedding search, which we did not attempt to scale beyond $n = 8$ (the held-fixed system size); a greedy or SABRE-style initial placement would be needed for larger lattices.

6 Conclusion

[Summary.]

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